Vertebrate Genome (VGP) Assembly Using Galaxy

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VGP Genome Assembly Pipeline:

1. **Input Data:** HiFi + Hi-C + Bionano

2. **Read Cleaning:** Adapter removal

3. **K-mer Analysis:** Genome estimation

4. **Assembly:** Hifiasm (graph-based assembly)

5. **Scaffolding:** Hi-C + Optical maps

6. **Validation:** BUSCO + Merqury

7. **Output:** Chromosome-level genome

Abstract

This study applies the Vertebrate Genome Project (VGP) assembly pipeline using the Galaxy platform to produce a high-quality, chromosome-scale genome of ***S. cerevisiae* S288C**. By combining PacBio HiFi reads, Hi-C interaction data, and Bionano optical mapping, the workflow generates a highly continuous and phased assembly. Multiple quality checks, including k-mer analysis and validation steps, confirm the accuracy and completeness of the final genome.

1. Introduction

Genome assembly reconstructs an organism's DNA sequence from raw sequencing data. Traditional short-read methods often struggle with repetitive and complex regions. The VGP approach overcomes these issues by integrating:

- HiFi long reads for high base accuracy
- Hi-C sequencing for chromosome-level organization
- Bionano optical mapping for structural validation

This project focuses on *Saccharomyces cerevisiae* (strain S288C), a model organism with a genome size of about 11.7 Mb across 16 chromosomes.

2. Materials and Input Data

Three types of datasets were used:

2.1 HiFi Reads

- Generated using PacBio SMRT sequencing
- Long reads (~10–25 kb) with very high accuracy (>99%)
- Used as the main input for genome assembly

2.2 Hi-C Reads

- Produced using Illumina paired-end sequencing
- Provide 3D chromatin interaction data
- Help in scaffolding and haplotype phasing

2.3 Bionano Optical Maps

- Based on labeled long DNA molecules
- Support large-scale structural organization and validation

All analyses were carried out on the Galaxy platform (usegalaxy.eu), ensuring a reproducible and user-friendly workflow.

3. Methodology

3.1 Data Preprocessing

HiFi reads were checked for adapter contamination using Cutadapt. Instead of trimming, reads containing adapters were completely removed to maintain assembly accuracy.
Outcome: Clean, high-quality reads for further analysis.

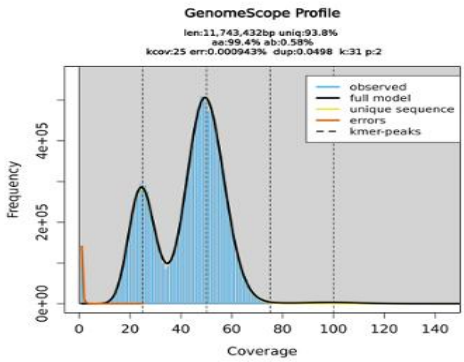
3.2 Genome Characterization

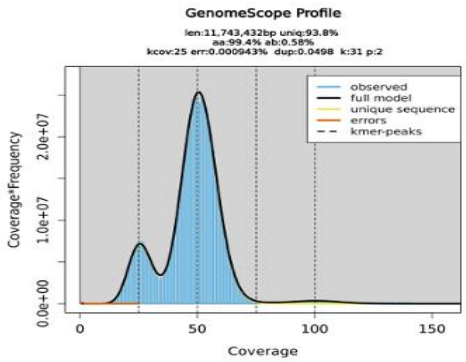
K-Mer Counting (Meryl):
31-length k-Mers were counted to generate a frequency distribution of sequence patterns.

Genome Modelling (GenomeScope2):
The k-Mer data was used to estimate genome features:

- Genome size: ~11.7 Mb
- Heterozygosity: ~0.58%
- Unique content: ~93.8%
- Error rate: very low

Interpretation: The genome is mostly homozygous with few repeats, making it suitable for accurate assembly.





3.3 De Novo Genome Assembly

Assembler: Hifiasm
Hifiasm builds a graph-based genome assembly and separates haplotypes using Hi-C data.

Process:

1. Fix sequencing errors
2. Build assembly graph
3. Split heterozygous regions into two haplotypes

Outputs:

- Hap1 contigs
- Hap2 contigs

The results are initially stored in GFA format, preserving sequence relationships.

3.4 Assembly Conversion and Statistics

Tool: Gfastats
Assembly graphs were converted to FASTA format for further analysis.

Key Metrics:

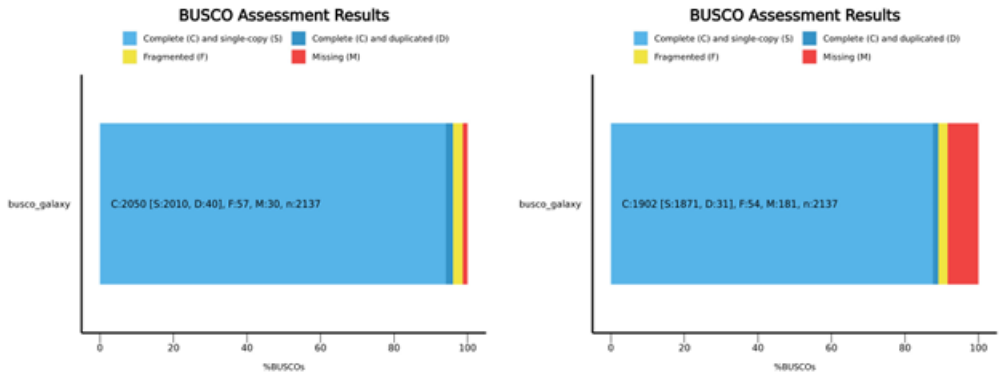
- Contigs: 16–17
- Largest contig: ~1.5 Mb
- N50: ~922 kb

Interpretation:
The assembly shows high continuity and aligns well with the expected chromosome number.

3.5 Assembly Completeness Assessment (BUSCO)

- **Hap1:** 95.9% complete, 1.4% missing, ~1.5% duplicated → **high-quality**
- **Hap2:** 89.0% complete, 8.5% missing, ~1.4% duplicated → **moderate quality**

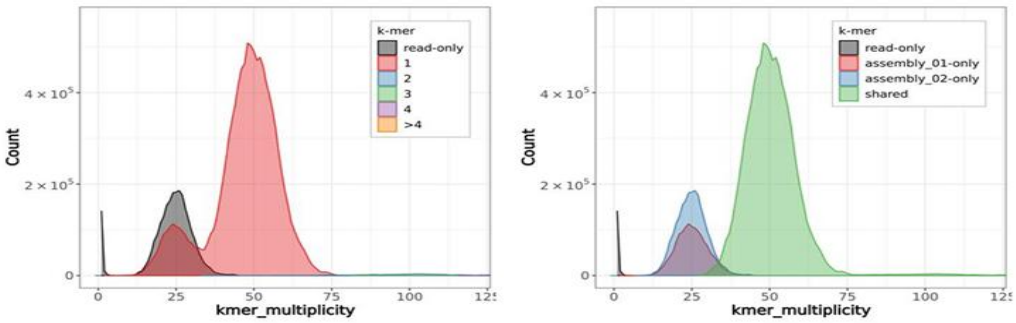
Hap1 is more complete; Hap2 is slightly less complete, consistent with diploid assemblies.



3.6 Reference-Free Quality Assessment (Merqury)

- **Combined completeness:** 99.9999%
- **Haplotype separation:** Distinct and well-resolved

Assembly is nearly complete and accurately phased, with minimal errors.



3.7 Structural Scaffolding

Bionano Hybrid Scaffolding

- Larger scaffold structures
- Improved genome organization

Hi-C Based Scaffolding (YaHS)

- **Process:** Map Hi-C reads → detect interactions → assemble chromosomes
- **Result:** Chromosome-level scaffolds

3.8 Validation Using Contact Maps

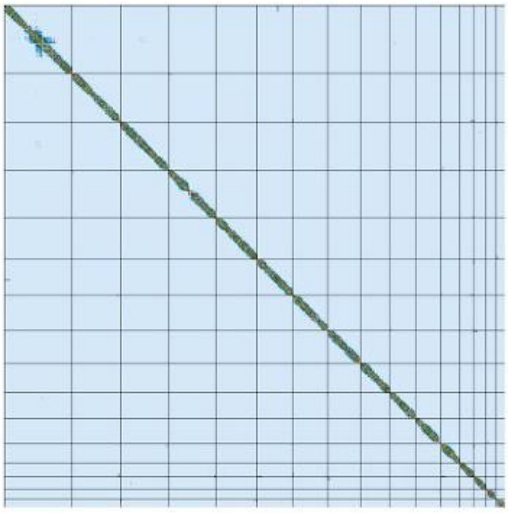
Before Scaffolding

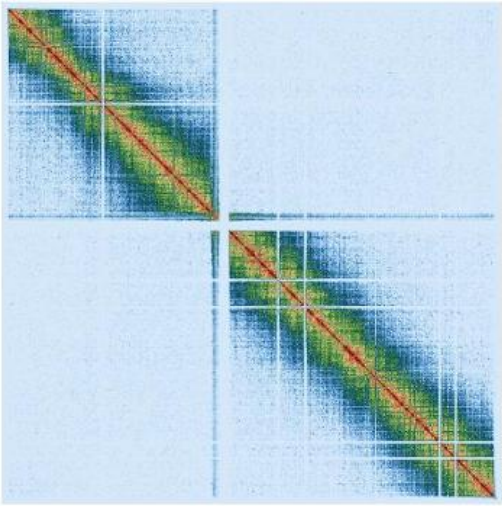
- Thin diagonal → contigs correct but unorganized

After Scaffolding

- Clear block patterns → chromosome-level structure

Successful chromosome-scale assembly with correct ordering and orientation.





4. Results and Discussion

4.1 Assembly Quality

- Near-complete genome reconstruction
- Chromosome-level scaffolding
- High contiguity and accuracy

4.2 Performance Evaluation

- **Genome size:** ~11.7 Mb → Accurate

- **N50:** ~922 kb → Excellent
- **BUSCO completeness:** 95.9% → High
- **Merqury completeness:** 99.9999% → Outstanding
- **Phasing quality:** Clear separation → Excellent

4.3 Biological Significance

- Accurate representation of both haplotypes
- Minimal duplication errors
- Reliable structural organization

5. Strengths of the Pipeline

- Integrates multiple complementary technologies
- Produces high-resolution genome assemblies
- Enables haplotype phasing
- Fully reproducible in Galaxy

6. Limitations

- Requires high computational resources
- Dependent on availability of multiple datasets
- Some manual curation may be required

7. Conclusion

- Near-complete genome assembly achieved
- High structural accuracy
- Correct haplotype resolution

Overall: A robust, chromosome-level assembly pipeline suitable for advanced genomic research.